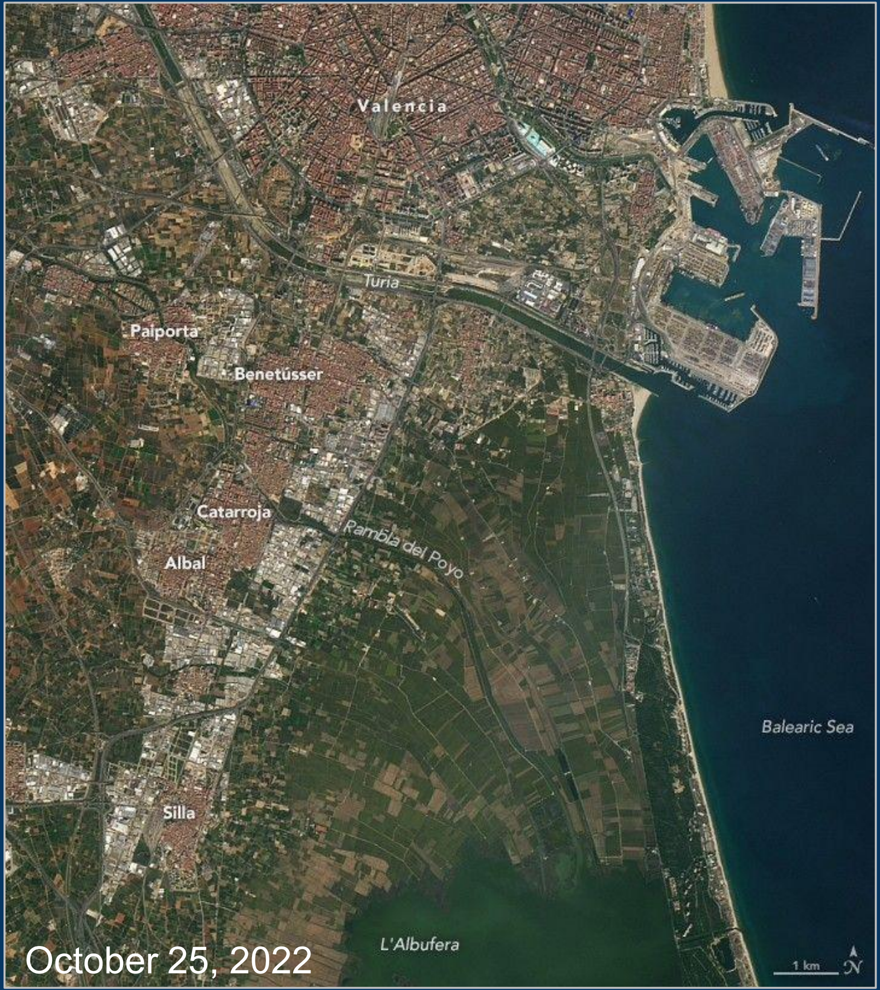
# Flood Monitor

WATER DEPTH PIPELINE

## Flood Exposure Analysis Across Space and Time

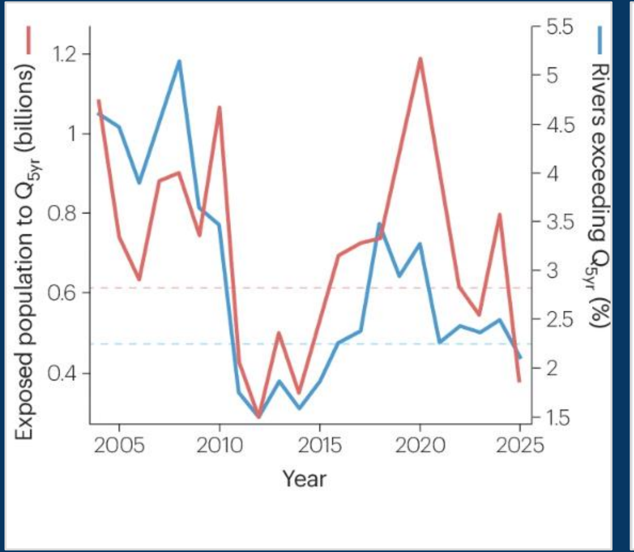




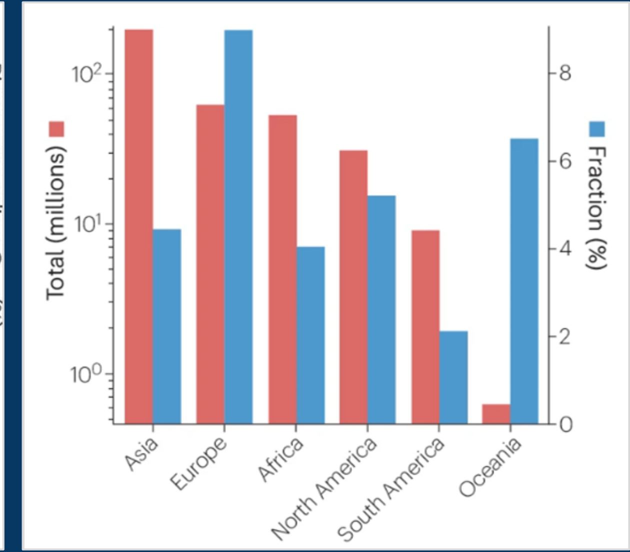
NASA, Lauren Dauphin, Landsat data from the USGS

# Floods Globally

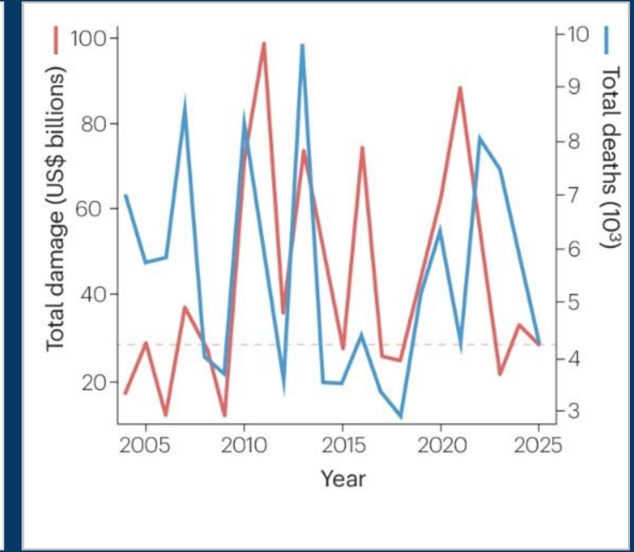
Global exposed population and rivers exceeding $Q_{5yr}$




Exposed Population to $Q_{5yr}$ in 2025




Annual flood-related deaths and damage (EM-DAT)




Getirana, A., Collins, E., Alcântara, E. *et al.* Floods in 2025. *Nat Rev Earth Environ* **7**, 270–273 (2026). https://doi.org/10.1038/s43017-026-00779-x

# The Problem

- Information about floods and elements at risk is fragmented across **different datasets**
- **Complex workflows** are required to turn geospatial data into operational flood intelligence to support response and planning
- Rapid mapping of flood extent and infrastructure exposure is **time critical**

# The Solution

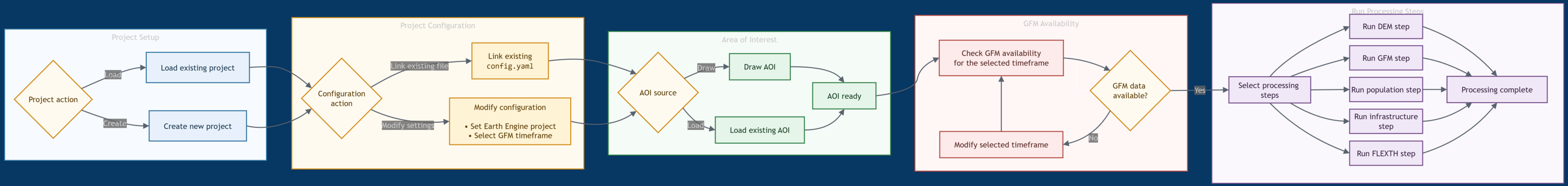
**Flood Monitor**
WATER DEPTH PIPELINE

- Unified pipeline to combine **flood data** and information about the **terrain** and **elements at risk** to calculate the **water depth** and **risk exposure across space and time**
- All packed into a clean **dashboard app**
- Easy-to-use → No geospatial data-science background required

# Flood Monitor App - Statement of Opportunity

- Supports **emergency response** and **local to global authorities**
- Fast and **near real-time** information about flooded areas
- **Cloud Computing** allows for fast computing without own infrastructure
- **Consistent workflow** across regions and events

# How it works (User Perspective)



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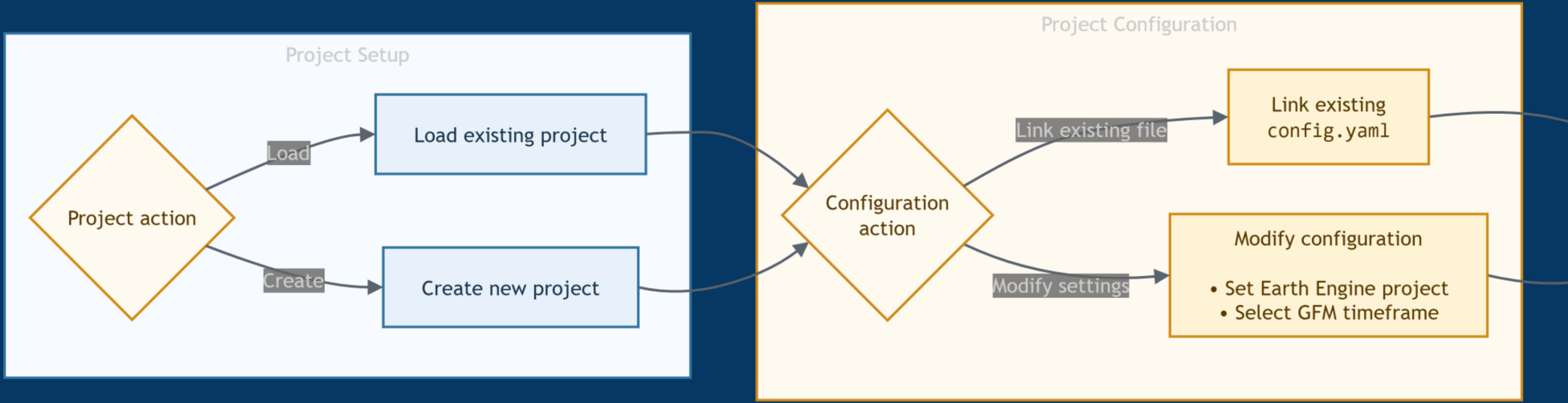

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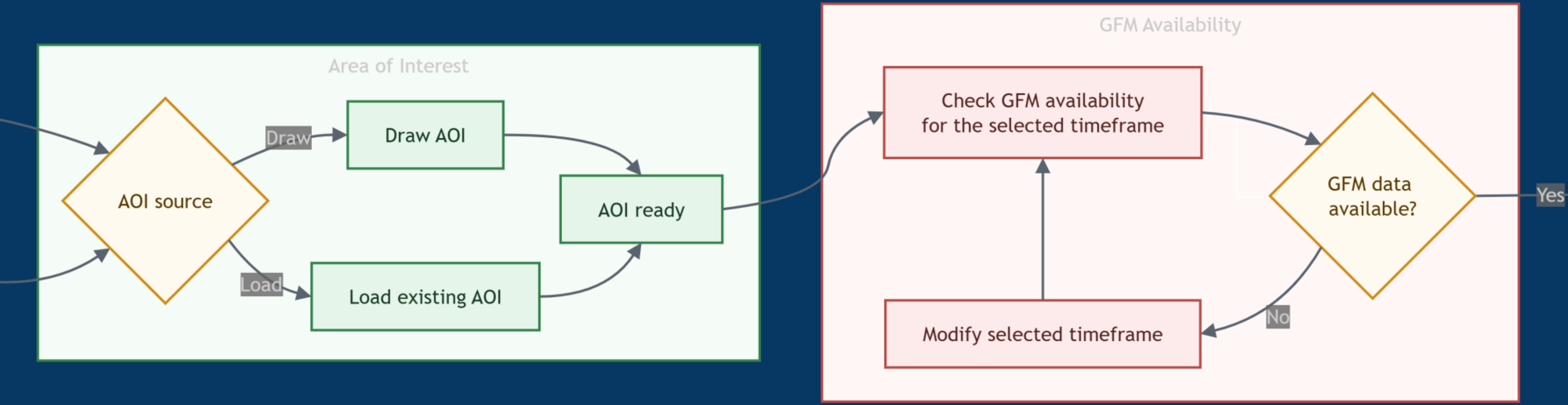

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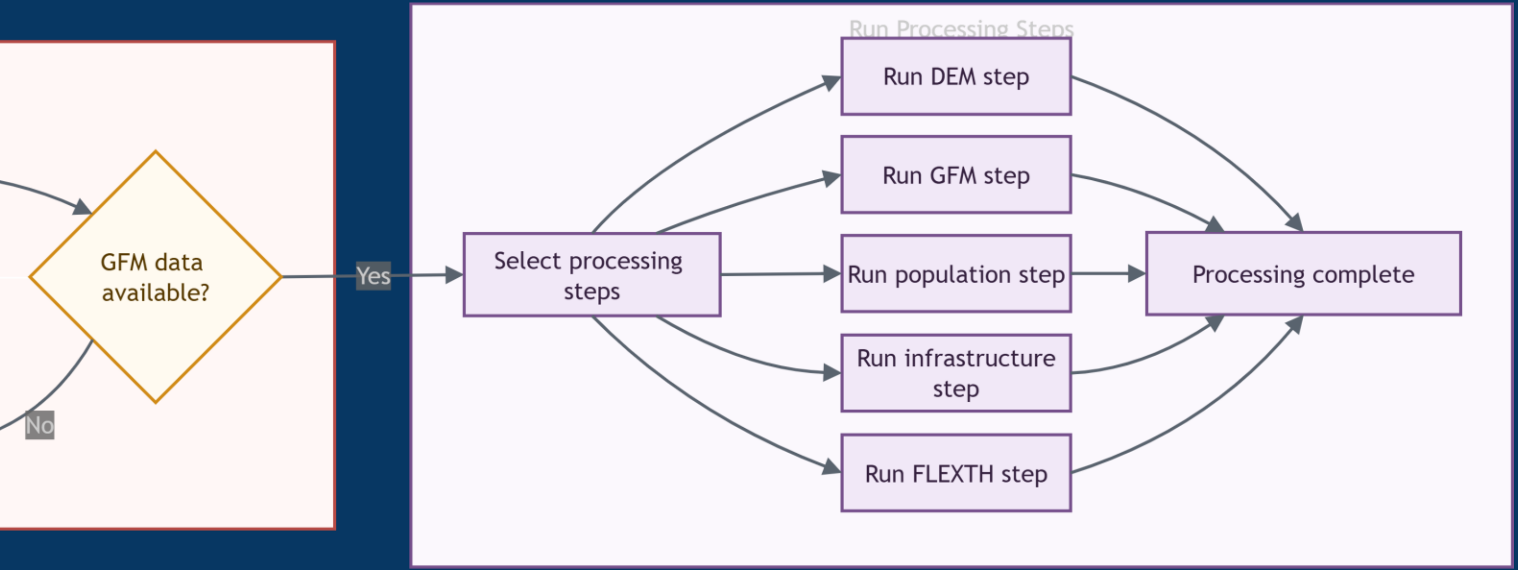
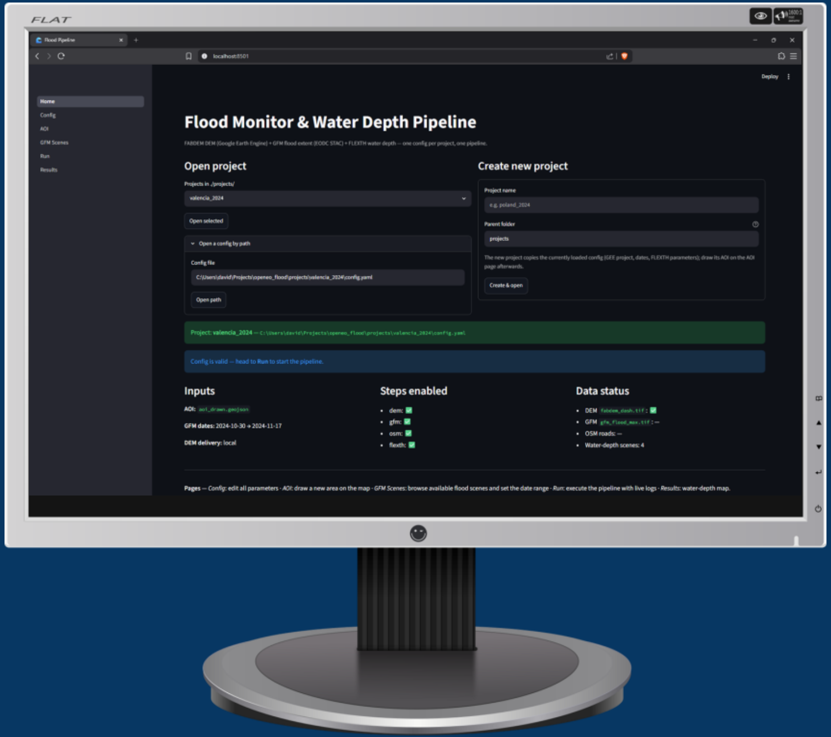


# Live-Demo


FLAT
Flood Pipeline
localhost:8501
Deploy
Home
Config
AOI
GFM Scenes
Run
Results
Flood Monitor & Water Depth Pipeline
FABDEM DEM (Google Earth Engine) + GFM flood extent (EODC STAC) + FLEXTH water depth — one config per project, one pipeline.
Open project
Projects in ./projects/
valencia_2024
Open selected
Open a config by path
Config file
C:\Users\david\Projects\openeo_flood\projects\valencia_2024\config.yaml
Open path
Create new project
Project name
e.g. poland_2024
Parent folder
projects
The new project copies the currently loaded config (GEE project, dates, FLEXTH parameters); draw its AOI on the AOI page afterwards.
Create & open
Project: valencia_2024 — C:\Users\david\Projects\openeo_flood\projects\valencia_2024\config.yaml
Config is valid — head to Run to start the pipeline.
Inputs
AOI: aoi_drawn.geojson
GFM dates: 2024-10-30 → 2024-11-17
DEM delivery: local
Steps enabled
dem:
gfm:
osm:
flexth:
Data status
DEM fabdem_dash.tif :
GFM gfm_flood_max.tif : —
OSM roads: —
Water-depth scenes: 4
Pages — Config: edit all parameters · AOI: draw a new area on the map · GFM Scenes: browse available flood scenes and set the date range · Run: execute the pipeline with live logs · Results: water-depth map.